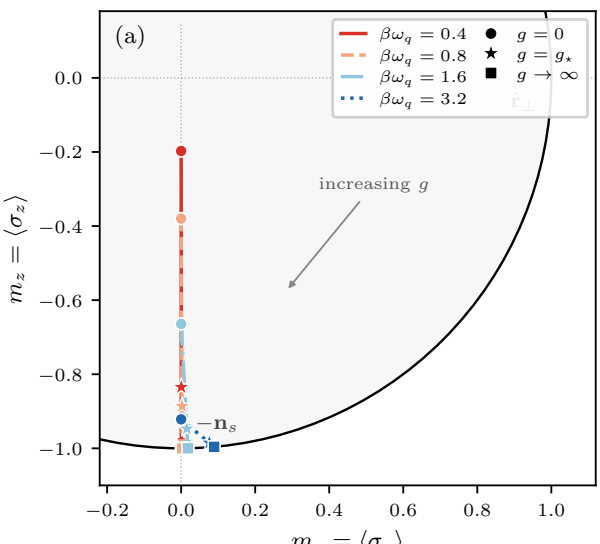
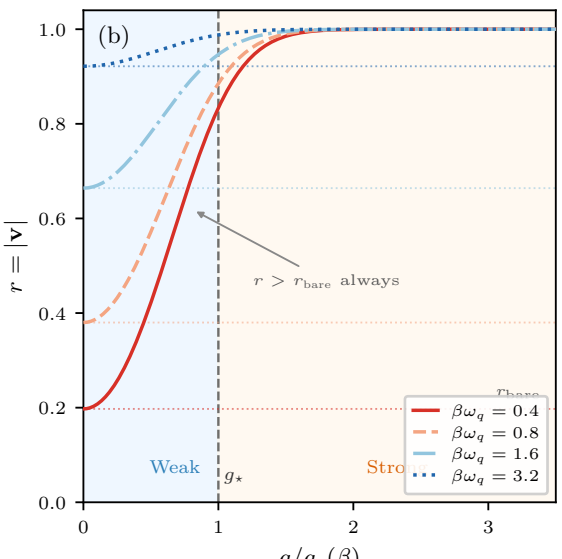

(a)
$\beta\omega_q = 0.4$
$\beta\omega_q = 0.8$
$\beta\omega_q = 1.6$
$\beta\omega_q = 3.2$
$g = 0$
$g = g_\star$
$g \to \infty$
increasing g
$-\mathbf{n}_s$
$m_z = \langle \sigma_z \rangle$
$m_x = \langle \sigma_x \rangle$


(b)
$r > r_{\text{bare}}$ always
Weak
$g_\star$
$\beta\omega_q = 0.4$
$\beta\omega_q = 0.8$
$\beta\omega_q = 1.6$
$\beta\omega_q = 3.2$
$r = |\mathbf{v}|$
$g/g_\star(\beta)$